Chapter 5
GAME THEORY

5-1. The Basic Concepts of Game Theory

Individual decision theory focuses on decisions involving the choices of one agent and outcomes determined by the agent's choices and the background environment. Its paradigms are those of a single person deciding whether to wear a raincoat, place a bet, invest in a stock, or attend a law school. Some of its examples, such as the choice between taking a course under Professor Smith rather than under Professor Jones, involve other individuals, but in analyzing such examples it treats the additional individuals as part of the environment. Now we will expand our analytic framework to include cases in which the decisions of other individuals are represented as playing an *active* role in determining the outcomes; these situations are called *games*. Games include ordinary parlor games, such as chess, bridge, Monopoly, or Scrabble; they also include serious decision-making situations as well, such as competitive bidding on oil leases, choosing weapons systems, pricing perfume, selling or holding stocks of grain, or proposing terms of peace.

To get a firmer grip on the game concept let us look closely at two games, a diverting one—chess—and a deadly serious one—negotiating a peace treaty. In chess there are two *players*; at a peace conference each nation is a player and there are often more than two. Each chess player aims to win; each nation negotiating peace seeks terms most favorable to it. Chess is played by one player making a move, the other responding, and so forth, until one wins or there is a draw. At the negotiating table players make proposals and counterproposals, but the outcomes are nowhere as clearly defined as in chess. Peace talks may drag on for years, with the attendant delays costing one or both sides dearly in human or economic tolls; or they may quickly conclude in a settlement because one nation has gained clear military superiority; or they may be broken off and the war continued. Furthermore, in real peace talks the rules are minimal and quite vague; there is no limited set of choices open to each negotiator nor are the outcomes limited to a definite set. In treating peace talks by means of game theory, however, we must assume that definite sets of choices are open to the negotiators, that the outcomes are limited, and that the "rules" determine which

outcomes are associated with each series of choices by the players. Unlike chess, the environment determines outcomes of peace talks too. If the rainy season is especially severe, the more mobile army can get the upper hand; if there are storms at sea, supply lines can be cut. In this respect, peace talks are best treated like gin rummy, bridge, poker, and other *games of chance*. This is done in game theory by introducing an additional player, called *Chance*, who makes choices using a randomizing device of one kind or another. Rolling dice in Monopoly, dealing hands of cards in bridge, dealing and drawing cards in gin are treated as moves by Chance.

Some of the elements of a game are beginning to emerge from our discussion. First, each game has two or more *players*. If the game is one of chance, there are at least three players with Chance being one of them. Second, each game has one or more *moves* with one or more players being allowed a choice at each move. Third, the rules of the game determine for each sequence of moves whether it results in an *outcome*, and if it does, what the outcome is. A sequence of moves terminating in an outcome of a game is called a *play*. Some games have short plays. The children's game scissors, paper, and stone is usually played with both players revealing their choices at the same time. The outcome is immediate: Scissors beats paper, paper beats stone, stone beats scissors, anything else is a draw. Chess, by contrast, has plays of every finite length, provided players are allowed to repeat sequences of moves.

Game theory uses two means for representing games formally. The *game-tree method* uses structures similar to decision trees to represent games. Each move is represented by a labeled node of the tree—a circle with a numeral within it—indicating which player is given a choice at that node. The choices open to the player are represented by branches emanating from the node. Outcomes are assigned to the end of each branch and are usually represented by means of one or more numbers that indicate the utility received by each player for the outcome.

A game tree for scissors, paper, and stone cannot represent the players as choosing simultaneously, because only one player is assigned to each node. Game theory handles this problem by representing the second player as making a choice without knowing what choice the first player made. (Similarly, the effect of simultaneous bidding on a contract is often achieved by having potential contractors submit sealed bids prior to an announced time at which all bids are to be opened.) Thus after the first player has made a choice, the second player will not know whether he or she is at the node following the first's choice of scissors or at the one following the choice of paper or the one following the choice of stone. These nodes are said to belong to the same *information set*, and the second player is said to have *imperfect information* because he or she is unable to distinguish between the various nodes within the information set. In the game tree depicted in figure 5-1, there is a dotted line surrounding all the second player's nodes to indicate that they all belong to the same information set. The outcomes are given as wins, losses, or draws for player 1.

Scissors, paper, and stone, as represented here, and bridge and Monopoly, in actual practice, are games of imperfect information: Even players with perfect

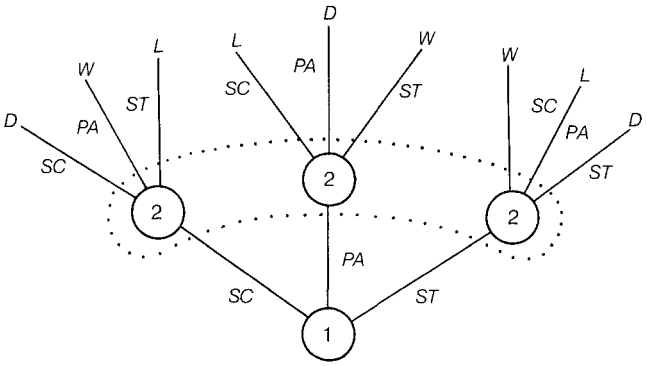

Figure 5-1

memories cannot know the entire history of the play preceding their moves. Thus, although in principle Monopoly players can know the assets and liabilities of the other players, and where they have landed, they cannot know, at least not until quite late in the game, what move Chance made initially when arranging the "chance" and "community chest" cards. Chess and checkers, on the other hand, are games of perfect information. Games of chance are not necessarily games of imperfect information; Monopoly without "chance" and "community chest" cards would be a game of perfect information despite Chance's making every other move. Nor are Chance moves required for a game of imperfect information; the tree version of scissors, paper, and stone has no place for Chance.

Game theory abstracts from most of the psychological and practical features of a game; it counts any two games having the same game tree as the same game. Chess played with carved ivory pieces is the same game as chess played with plastic pieces. That is not remarkable. On the other hand, it is interesting that chess played with a one-second time limit is still regular chess as far as game theory is concerned. The next example is even more notable. The game of fifteen has two players who use nine cards, an ace through a nine. These cards are laid face up on a table in front of the players, who take turns picking up cards. The object of the game is to achieve a total of 15 with exactly three cards before the other player does. (An ace counts as a 1.) The game ends in a draw if all the cards are drawn with no player achieving a 15 with exactly three cards. Now consider any tick-tack-toe board and think of the squares as numbered according to the scheme in table 5-1.

The diagonals, the middle column, and the middle row sum to 15; thus the game of fifteen has the same game tree as that restricted version of tick-tack-toe that counts "three in a row" only if they fall on a diagonal or middle row

or column. If you draw game trees for restricted tick-tack-toe and the game of fifteen you will see that they are the same. (Do not try unless you have a big

5-1

1	2	3
4	5	6
7	8	9

sheet of paper, time, and patience; the first move has nine branches.)

Just as decision trees can be reduced to decision tables, so too can game trees be reduced to game matrices or tables by using strategies. Two-person games can be represented by m-by-n tables, with the m rows and n columns corresponding to strategies and the entries in the squares representing the *payoffs* or outcomes for each player for the pair of strategies (row, column) determining the square in question. Here is game table 5-2 for scissors, paper, and stone.

5-2

		Col's Strategies		
		SC	*PA*	*ST*
Row's Strategies	*SC*	*D, D*	*W, L*	*L, W*
	PA	*L, W*	*D, D*	*W, L*
	ST	*W, L*	*L, W*	*D, D*

The first payoff given in each square is for the player using the strategies assigned to the rows; the second payoff is for the player using those heading the columns. For convenience we will call the first player "Row" and the second "Col." Thus when Row plays *SC* (scissors) and Col plays *ST* (stone), the payoff to Row is an *L* (loss) and that to Col is a *W* (win). Any game involving a series of moves will have a large number of strategies and, thus, a huge table or matrix. Despite this, many of the theoretical and philosophical features of game theory can be best understood in terms of games in matrix form. Consequently we will henceforth restrict ourselves to games in this form.

Chance is no longer represented in the matrix form of a game. Her role is absorbed in the payoffs. The example of gin will be used to explain how this happens. Gin is put into matrix form by having both players make just one move each—they pick a strategy. Their individual strategies will consist in a complete list of instructions for reacting to any hand they may be dealt, any cards they may draw, any cards their opponent may discard, and so on. Because gin is a game of chance, a selection of strategies by players alone does not determine a unique outcome; different distributions of the cards will generally result in different winners for the same selection of strategies. Yet each selection of

strategies does determine a lottery among the possible outcomes, with the possible distributions of cards yielding the probabilities in the lottery. Suppose, for example, the game is being played by just Row and Col, who have selected two strategies R_1 and C_1. Let us further suppose that for a third of the arrangements of the cards the pair (R_1, C_1) yields a win for Row, for another third it yields a win for Col, and for the balance it yields a draw. Then we can associate with the pair (R_1, C_1) the lottery that yields an equal chance of a draw, Row's winning, and Col's winning. This takes Chance's role as a player and adsorbs it into the outcomes.

Game theory takes this a step further. Each player is assumed to have a utility function for outcomes that abides by the expected utility theorem. Thus the utility of an outcome that happens to be a lottery is equal to its expected utility, and outcomes, whether lotteries or certainties, can be represented by utility numbers. The net effect is that a game of chance and a game of certainty could be represented by the same game matrix and would thus be counted as the same game by the theory. Here is an artificial example to illustrate this. In the game of even and odd two players choose between two strategies, E and O. E beats O; otherwise there is a draw. If we use 1 to represent a win, -1 a loss, and 0 a draw, the table for the game is given on the left (5-3). (Payoffs are given

5-3

	E	O
E	0	1
O	-1	0

5-4

	E	O
E	0	0
O	0	0

for Row.) Consider now a chancy version of even and odd. Players again choose between the two strategies, but then a coin is flipped. On heads, the result is the same as in even and odd; on tails, the result is calculated as if each player had the opposite strategies. Thus the pairs (E, E) and (O, O) yield the same results in either game, since each ends in a draw. Yet in the chancy version the payoff to Row for (E, O) is a coin toss, which yields a win on heads and a loss on tails. Thus its expected utility is $(1/2)1 + (1/2) - 1$, namely, 0. Finally, consider a two-player game of certainty with two strategies that always ends in a draw. This will have the same game table as the chancy version of odd and even; namely, table 5-4.

As we have already seen, game theory assumes that the players have utility functions producing interval scales for outcomes and lotteries involving outcomes. The theory also assumes that given a choice of strategies by his opponents, each player will choose those strategies that maximize his expected utility. On the other hand, the theory does not require the players to prefer winning; if a parent prefers to throw a game of checkers to boost her child's spirits, the parent's utility function will assign higher utility to losing. However, game theory will represent this version of checkers by a different table or tree than the standard version, since high utilities for winning outcomes will be replaced with

low ones. Despite this, whichever game the parent plays, the theory will assume that she seeks to maximize her utility.

Game theory also assumes that all players know both the entire game tree or table, whichever is relevant, and their own and all other players' utilities for outcomes. Furthermore, each player is taken to know that every player knows all this. This means that in thinking about a game all players can assume that they are playing with perfect opponents, who will always make optimal responses to every move or strategy choice. *The object of game theory is to determine the outcome or possible outcomes of each game, given these assumptions about the players.* To do this for a game is to *solve* it.

Before proceeding further let us see how game theory solves a simple game. The game is given in table 5-5. The best strategy for Col is C_2; he re-

5-5

	C_1	C_2	C_3
R_1	0, 0	1, 2	0, 2
R_2	1, 3	1, 4	0, 0

ceives 2 or 4, which is at least as good as his payoffs for C_1 and C_3. The same reasoning shows that the best choice for Row is R_2. Thus the game will be played via the strategy pair (R_2, C_2); Row will get a payoff of 1, Col one of 4. (Note: Row's utilities are taken to be fully specified; it would be fallacious to reason that Row would prefer to see Col get 2, and thus, knowing that C_2 will be chosen in any case, will choose R_1.)

There are several branches of game theory. The most important division is between the theory of two-person games and that of games involving more than two persons. (In this classification Chance does not count as a person.) The latter is usually called "n-person game theory." It is much more complicated than the two-person branch, because players can form coalitions and overwhelm another player, which makes it useful in analyzing political and economic situations. Unfortunately, the n-person theory is also in a less satisfactory mathematical state than the two-person theory. Accordingly, we will restrict our discussion, in the main, to the two-person theory.

Games are also classified according to whether the players' interests are entirely opposed (strictly competitive games) or in partial agreement. Monopoly and chess count as strictly competitive, but two drivers trying to avoid a head-on collision are involved in a game that is not strictly competitive. Nonstrictly competitive games are further divided according to whether the players are permitted to communicate with each other to coordinate stategies. If they can, the game is called a *cooperative game*. These games may allow the players to form binding agreements and even to redistribute the payoffs. Labor negotiators play cooperative games that end in binding agreements. When a play is sold to a movie company, the playwright's agent and a movie producer play a cooperative game which ends with a contract for distributing the profits of the sale. On the

other hand, two friends who have been separated at a crowded rock concert play a simple nonstrictly competitive game as they try to find each other.

Parlor games and sports may be the only games that are strictly competitive in reality. Even in war no nation seeks the total destruction of its opponent; thus the players have some common interests, and at peace conferences nations not only communicate but also agree to share certain payoffs. Despite this, game theorists often find it useful to analyze wars as strictly competitive games, since in the choice of strategies for battle the noncompetitive aspects of war are usually negligible.

In a strictly competitive game communication and contracts play no role, even if the rules permit them. For there is no reason to trust one's opponents; they can be expected to lie or cheat if doing so promotes their interests. Every player knows this and thus puts no stock in what the others say, threaten, or gesture. Game theory reflects this by letting lapse the distinction between games with and without communication when analyzing strictly competitive games.

PROBLEMS

1. Which of the following are games of perfect information?
   a. checkers
   b. poker
   c. tick-tack-toe
   d. bridge
   e. Chinese checkers
2. Discuss basketball from the point of view of game theory. How many players are there? Is an actual basketball game a single big game or a series of games? Are the games strictly competitive or cooperative? Does communication play any role?
3. Discuss the process of nominating a candidate for the U.S. presidency in game theoretic terms. Who are the players? Is the game strictly competitive? Can the players make agreements to distribute payoffs?
4. Why does prohibiting communication make no difference in the outcome of a strictly competitive game among perfectly rational players?

5-2. Two-Person Strictly Competitive Games

The most fully developed part of game theory is the theory of two-person strictly competitive games. Since in many ways it is also the easiest portion of game theory to grasp, we will start our study with it.

In a strictly competitive game the preferences of the two players are in exact opposition. To the extent one gains from an outcome of the game, the other loses. If we think of the gains and losses in terms of monetary payoffs, the winner receives the winnings in terms of a payoff from the loser. But the very reasons that led us to replace monetary values with utilities in individual decision theory will again lead us to make the same substitution in game theory. First, monetary or other concrete payoffs might not reflect the true values of the players. In a game between two unmitigated altruists, winning in the ordinary sense is actually the worst outcome for each. Yet their preferences are in com-

plete opposition: For them the game is strictly competitive—each does his damnedest to lose—but their preferences are just the opposite of those of ordinary players. For that reason they are playing a different game, and thus representing it requires us to replace concrete payoffs with the utilities each player assigns them. A second reason for using utilities is that outcomes frequently are best represented as lotteries. When this is done, explaining why maximizing expected utilities is the rational approach to these games—even when played only once or just a few times—drives us to identify the utility of a gamble with its expected utility. The need for doing this will be even more pressing later when we introduce mixed strategies.

In any two-person strictly competitive game the utility function of Col is the negative of that for Row, since Row and Col have strictly opposing preferences. Let $u(A)$ be Row's utility for the outcome A. Then Col's utility for A is $-u(A)$, and their utilities sum to zero. For this reason two-person strictly competitive games are often known as *zero sum games*. In a zero sum game table or tree we need not present the utilities for both players, since one's can be deduced from the other's. From now on payoffs will be given in the form of Row's utilities only.

Table 5-6 presents a simple zero sum game. Let us see how the theory solves it. Remember, Row seeks to maximize the outcome numbers and Col

5-6	C_1	C_2	C_3	C_4
R_1	0	1	7	7
R_2	4	1	2	10
R_3	3	1	0	25
R_4	0	0	7	10

tries to keep them down. A quick check will show that none of Row's strategies are dominated by the others, and that Col's C_4 is dominated by all his other strategies. So Col will not play C_4 and Row knows this. Knowing this, he also knows he can now ignore all the entries in the fourth column when picking his strategy. The game, in effect, reduces from a four-by-four game to a four-by-three game. But then R_3 can be deleted; it is dominated by R_2 once the 25 at the end of the row is ignored. Our game now amounts to this one (table 5-7).

5-7	C_1	C_2	C_3
R_1	0	1	7
R_2	4	1	2
R_3	0	0	7

In this game, however, C_3 is dominated by both C_1 and C_2; so Col will not play it. Knowing this, Row will play R_2, and knowing that in turn, Col will play C_2. So the solution to the games is this: Row and Col play the strategy pair (R_2, C_2), and the payoff to Row is 1. The payoff received by Row according to the game's solution is called *the value of the game.*

PROBLEMS

Use dominance reasoning to solve the following games (tables 5-8, 5-9, 5-10).

5-8

7	6	4
3	2	1
5	6	4

5-9

10	1	3
4	2	4

5-10

2	2	4	5
7	1	5	3
4	2	3	1

5-3. Equilibrium Strategy Pairs

One might think that all zero sum games can be solved by simple considerations of dominance—if they have a solution at all. But table 5-11 shows that this plainly is not so. Neither Row nor Col has a dominated strategy; so the process of elimination used to solve the last game cannot get started. Despite that, the

5-11

	C_1	C_2	C_3
R_1	8	8	7
R_2	0	10	4
R_3	9	0	1

solution of the game is readily at hand. Consider the strategy pair (R_1, C_3). If Row knew that Col would play C_3, then he, Row, would play R_1 because that is the best he can do against C_3. On the other hand, if Col knew that Row would play R_1, then he, Col, would play C_3 since that would bring Row's payoff down to the lowest point possible. The strategies (R_1, C_3) are in *equilibrium* in the sense that once this pair is chosen neither player will prefer to change unilaterally to another strategy. He *cannot* do better by changing. In general we will say that a pair of strategies is an *equilibrium pair* or is in equilibrium just in case

neither player can do better by unilaterally changing his strategy. The payoff associated with an equilibrium pair is called an *equilibrium value.*

Once the players hit on an equilibrium pair, neither has any reason for changing his strategy choice unless the other player does too. But what reason is there to expect that they will hit on one? In a zero sum game communication and bargaining are excluded; the only way they could both decide upon an equilibrium pair is for each to decide on his own that his half of the pair is the best strategy for him. How can that happen?

Return to the game presented in table 5-11. Other things being equal, the best strategy for Row is R_2; it gives him a chance at the highest payoff. But other things are not equal; Row must contend with Col's strategy choice, and Col would pick C_1 if he thought Row would pick R_2. Row knows this, and he knows that Col knows that he knows, and so on. On the other hand, Col would most prefer the payoffs of 0 afforded by C_1 and C_2 – if Row would only cooperate and pick R_2 and R_3, respectively. But Col knows that he cannot count on that. Moreover, Row can see that if he picks R_1, he can depend on a payoff of at least 7, whereas any other strategy might yield a 0, a 1, or a 4. Col also sees that he can depend on holding Row down to a 7 if he, Col, chooses C_3, but any other choice might allow Row a gain of 8, 9, or 10. Each sees then that by playing his half of the equilibrium pair he can guarantee a certain value, *his security level* for the game; playing another strategy might lead to a less favorable outcome. Given that each knows he is playing against a perfect player, neither will risk taking a strategy that does not yield his security level for the game. By independent reasoning, then, each picks his half of the equilibirium pair.

The definition of equilibrium pairs is fit for other branches of game theory. There is, however, a simpler criterion for equilibrium pairs in zero sum games. Let us call it *the minimax equilibrium test for zero sum games.* It reads: *In a zero sum game a necessary and sufficient condition for a pair of strategies to be in equilibrium is that the payoff determined by them equal the minimal value of its row and the maximal value of its column.* In the game in table 5-12, for example, the equilibrium test picks out (R_2, C_2) as the equilibrium pair. Notice that al-

5-12

	C_1	C_2	C_3
R_1	1	3	6
R_2	7	5	5
R_3	3	4	10

though the 5 at the intersection of row 2 and column 3 is the minimal value for row 2, it is not the maximal value for column 3; so the pair (R_2, C_3) is not in equilibrium.

Some games have several equilibrium pairs. The game in table 5-13 has two, that in table 5-14 has four. This raises a serious question: If a game has

more than one equilibrium pair, need the players worry about coordinating their choices to guarantee that the strategy pair they pick will in fact be in equilibrium? Since communication, and thus coordination, is pointless in a zero sum game, the use of equilibrium solutions for zero sum games appears to be jeopardized. Fortunately there is no need to worry. The answer to our question is negative. Examine the two games in tables 5-13 and 5-14. In the first, Row

5-13

	C_1	C_2	C_3
R_1	0	8	3
R_2	0	1	10
R_3	−2	6	5

5-14

	C_1	C_2	C_3	C_4
R_1	1	2	3	1
R_2	0	5	0	0
R_3	1	6	4	1

has to worry about coordination; his equilibrium strategies are R_1 and R_2. But this is no problem for Row; for both (R_1, C_1) and (R_2, C_1) are in equilibrium. Either choice he makes will be fine. The second example is more complicated. Row has two choices, R_1 and R_3, and Col has two also, C_1 and C_4. Yet no matter which choice either makes they will achieve equilibrium; (R_1, C_1), (R_1, C_4), (R_3, C_1), and (R_3, C_4) are all equilibrium pairs and each has the same value.

These examples are instances of a general theorem, which I will call the *coordination theorem for zero sum games*. It reads:

> *If, in a zero sum game, the pairs (R_i, C_j) and (R_l, C_m) are in equilibrium, so are the pairs (R_i, C_m) and (R_l, C_j); furthermore, the values for each of the pairs are the same.*
>
> PROOF. Let v_{ij}, v_{lm}, v_{im}, and v_{lj} be the values for the pairs (R_i, C_j), (R_l, C_m), (R_i, C_m), and (R_l, C_j), respectively. Since the game is assumed to be zero sum, the equilibrium test applies to it. Thus v_{ij} and v_{lm} are minimal values for their rows and maximal values for their columns. Since v_{ij} and v_{im} occur in the same row and v_{ij} is an equilibrium value, we know that
>
> (1) $v_{ij} \leq v_{im}$.

But since v_{lm} and v_{im} are in the same column and the latter is an equilibrium value, we have

(2) $v_{im} \leq v_{lm}$.

Since v_{lj} and v_{lm} are in the same row, we have

(3) $v_{lm} \leq v_{lj}$.

Finally, since v_{lj} and v_{ij} are in the same column,

(4) $v_{lm} \leq v_{ij}$.

Putting (1)–(4) together we find that

(5) $v_{ij} \leq v_{im} \leq v_{lm} \leq v_{lj} \leq v_{ij}$,